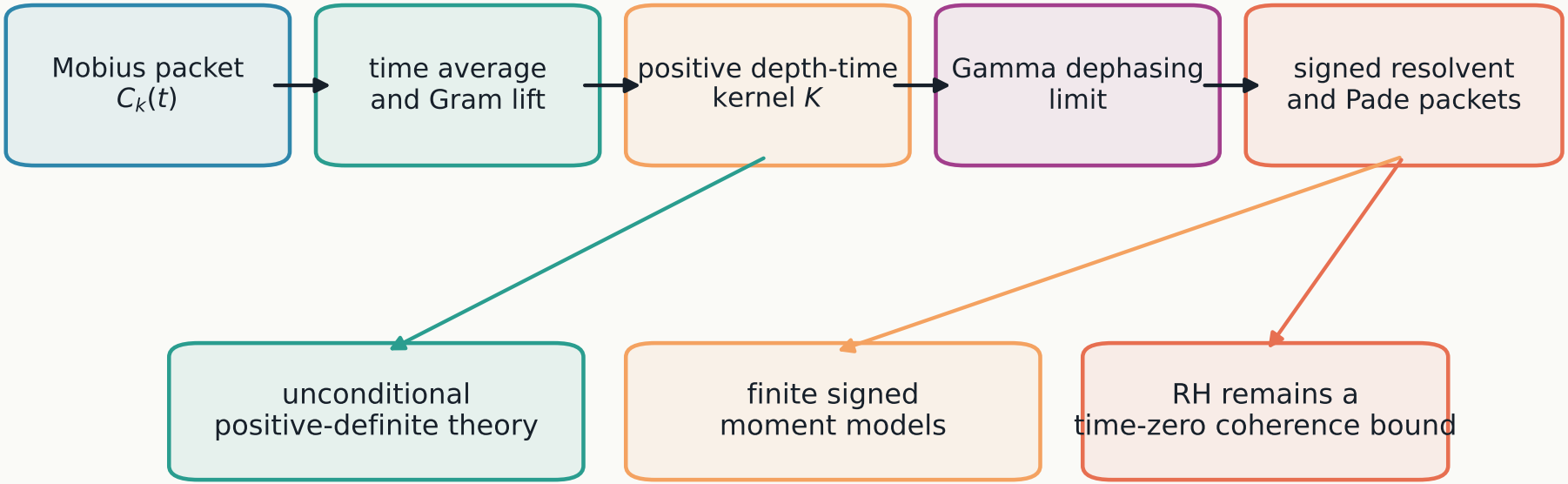From signed arithmetic transfer to universal depth-time kernels